COURSE SYLLABUS

Introduction to Engineering Design

2026–27 · The Cushman School

TEACHER
Mr. Willie Avendano

EMAIL
wavendano@cushmanschool.org

ROOM
Room 502/3

CLASS MEETS
Monday, Period 6 (12:42–1:26) · Wednesday & Friday, Block 6 (9:36–10:56)

EXTRA HELP
3:15–4:00 PM on Mondays, Tuesdays, and Wednesdays, or by appointment. *Dismissal runs 3:00–3:15, so help begins at 3:15. Thursdays are unavailable — HS faculty meeting.*

Course Description

PLTW Introduction to Engineering Design (IED) is a high school engineering course in the PLTW Engineering pathway. In IED, students explore engineering tools and apply a common approach to the solution of engineering problems: an engineering design process. Using the activity-, project-, and problem-based (APB) teaching and learning pedagogy, students progress from completing structured activities to solving open-ended projects and problems that require them to plan, document, communicate, and develop other professional skills.

Through both individual and collaborative team activities, projects, and problems, students apply systems thinking and consider various aspects of engineering design including material selection, human-centered design, manufacturability, assemblability, and sustainability. Students develop skills in technical representation and documentation, especially through 3D computer modeling using a computer-aided design (CAD) application. As part of the design process, students produce precise 3D-printed engineering prototypes using an additive manufacturing process. Student-developed testing protocols drive decision making and iterative design improvements.

To inform design and problem solutions addressed in IED, students apply computational methods to inform design by developing algorithms, performing statistical analyses, and developing mathematical models. Students build competency in professional engineering practices including project management, peer review, and environmental impact analysis as part of a collaborative design team. Ethical issues related to professional practice and product development are also presented.

Units

This course follows the official **PLTW Introduction to Engineering Design** units. Each is taught through the PLTW lessons listed beneath it.

Unit 1 **Design and Problem Solving**

- 1.1 Design Basics
- 1.2 Visualization and Solid Modeling
- 1.3 CAD Fundamentals
- 1.4 Product Improvement

Unit 2 **Assembly Design**

- 2.1 Put It Together
- 2.2 Take It Apart
- 2.3 A Material World
- 2.4 Fix It

Unit 3 **Thoughtful Product Design**

- 3.1 Responsible Design
- 3.2 More Than Parts
- 3.3 Solve a Problem

Unit 4 **Making Things Move**

- 4.1 You've Got to Move It
- 4.2 May the Force Be With You
- 4.3 Automating Motion
- 4.4 Make It Move

Week-by-week pacing — including which weeks carry no new content — is on the course page.

Required Texts and Links

Subject to change and grow.

- PLTW Engineering Online Student Textbook (via Clever)
- PLTW Engineering Notebook
- Autodesk Fusion 360 (free education license)

Course Materials

- School Laptop
- Internet Connection

- Computer Charger
- Pencil, Pen (mandatory for drawings)

Grades and Evaluations

Grades will be calculated using the 4.0 scale as mentioned in the student handbook.

70%
Summative
unit deliverables, portfolios, CAD models, design reports, presentations, and end-of-unit projects

25%
Formative
engineering notebook entries, in-class activities, sketching practice, guided CAD exercises, class discussions, and unannounced quizzes

5%
Diagnostics
Entrance and exit tickets, and other checks that are neither summative nor formative.

Description of Common Assessments

Weekly deliverables are the spine of this course. Something ships every Friday (Wednesday on short weeks) — sketch portfolios, modeled parts, analysis reports, prototypes. Instructions, requirements, materials, and rubrics are discussed beforehand and posted in Veracross.

The engineering notebook is a graded, ongoing record of your design work: sketches, calculations, decisions, and iterations, dated and in ink. It is audited at midterm and again at the end of the year. Professional engineers document as they work; so do we.

Projects — including the reverse-engineering study, the redesign, and the year-end design project — are extended, largely in-class efforts that may require work outside the classroom. These involve independent research, technical drawings, physical prototypes, and oral presentations.

Classwork (CW in Veracross) should be completed during the allotted class time and is **graded for accuracy**. Since these assessments are formative, students will have the chance to resubmit work within an allotted time determined by the teacher.

Homework (HW in Veracross) assignments allow students to practice their skills outside of the school setting or prepare themselves for the following class. Homework assignments are due at the start of the class period.

Scoring on homework will be as follows:

- **4** — All components of the assignment are completed in full, addressing all requirements with a high level of detail.
- **3** — Most components of the assignment are completed, meeting most of the requirements with satisfactory detail.
- **2** — Some components of the assignment are completed, but gaps or missing elements are noticeable. Detail is lacking in some areas.
- **1** — Few components of the assignment are completed, and significant gaps or missing elements are evident. Details are minimal.
- **NTI** — No assignment is turned in and is subject to the late work policy.

Rules and Routines

Your success in this course will directly reflect on the amount of time and energy you put into it.

Rules

Respect. This is the number one rule in the class. Respect you, your peers, and Mr. Avendano.

Consequences. Most violations will undergo the following consequences:

1. **1st Offense:** Verbal warning
2. **2nd Offense:** Conference at the end of class
3. **3rd Offense:** Behavior report to administration

Routines

1. **Class preparation.** Students are expected to come to class with materials ready, including and not limited to, having their laptops charged for class, with an adequate computer charger prepared to charge their computers if need be. I will have limited materials such as pens, pencils, calculators, and paper available. However, students who are always relying on these and do not consistently have their own materials will receive scores of 1 or 2 in responsibility (IRC).
2. **Participation.** There are many ways to participate in class and I expect students to partake in some way. This can include adding to class discussions, asking questions about class materials, working collaboratively during group work, attending extra help, etc.
3. **Absences.** Consistent attendance is strongly recommended as it facilitates the learning process. It is the student's responsibility to inform me via email or in person when they are or will be absent. Students will have two class period days to make up work from absences (Student Handbook, p. 21). Any work handed in afterwards will be subject to the "Late Work Policy."
4. **Food & Drinks.** Students may not eat or drink in the classroom. Water is the only exception. If you need to eat, you may briefly step outside when I allow permission. No visits to the vending machines are allowed.
5. **Late Work.** Students will have one week to submit any late work for partial credit. Late formative assessments can receive up to a 2.0 in the gradebook. It is not fair to give the same credit to students who turn their work in on time as to those who turn their work in late. However, if a student has an *extenuating circumstance*, I am more than willing to make accommodations. See me BEFORE the due date if this applies to you! **Late work will NOT receive credit after one week.** After the week, the highest possible grade is a 0. In this case, the student will receive a Complete/No Credit or an NTI if never turned in.
6. **Resubmissions/Test Corrections.** If a student obtains a score of 2 or lower on a written exam, they may request to complete test corrections only by filling out the necessary Resubmit Form. These are done only during extra help and in my presence. Students must arrange a time with me to complete corrections within a given deadline and will earn 1/2 credit back on their missed points. The original work must have been submitted on time to resubmit. Students who want to improve an unannounced quiz score may retake a similar quiz (once). This policy only applies to the aforementioned assignments. It does not apply to regular assignments, announced quizzes, or projects.
7. **Electronics.** Cell phones must be stored in the pouch, put away in the backpack, and on silent. Students are expected to use their laptops solely for academic purposes. A student who does not comply with this

expectation is not showing respect for themselves, their classmates, or their teacher and will be in violation of the classroom rule. Along with the application of consequences, the student will receive scores of 1 or 2 in industry, courtesy, and responsibility.

8. **Interaction with the Makerspace Area.** The classroom includes a makerspace area dedicated to hands-on activities like robotics, 3D printing, and electronics. This area is not to be accessed or used outside of the Engineering classes unless specified by the teacher for a particular project. Students must respect all makerspace equipment and materials, ensuring they are not tampered with, moved, or used for non-course-related activities.
9. **Restroom.** If a student needs to use the restroom, they may raise their hand and point at the door. When given permission, students may go. Due to the fact that class is on the fifth floor and the restroom is only permitted for faculty on the fifth, students are permitted to go to another floor for the restroom. The time used for the restroom must be kept to a minimum and may be subject to be tracked. Excessive use of the restroom will result in students using their passes for restroom use and/or parents and administration may be informed.

Academic Integrity

Students are expected and required to be honest in the preparation of assignments, papers, projects, and all other assessments. Cheating (including plagiarism) in any form is not allowed and it hinders the learning process. If a student chooses to do so, the work will receive a 0 and will lead to further disciplinary consequences. If a student cheats off a classmate's work, both will receive the above-mentioned outcomes.

Disclaimer

The institution and teacher reserve the right to make modifications to this information throughout the academic school year, should it be necessary.

Acknowledgement of Receipt and Understanding

Please sign and submit as soon as possible, but **no later than midnight on Sunday, August 30, 2026**.

> **Midnight means midnight.** Anything submitted at 12:01 AM Monday or later is late. Submissions are checked Monday morning.

I have read and fully understand the academic and behavioral expectations outlined in Mr. Avendano's Introduction to Engineering Design syllabus. I agree to abide by these guidelines or will undergo the academic and/or disciplinary consequences.

Student Printed Name ______________________________

Student Signature ______________________________

Parent/Guardian Printed Name ______________________________

Parent/Guardian Signature ______________________________

Date ______________________________